1- Masala

Berilgan

$v_1 = 90$ km/soat

$v_2 = 72$ km/soat

$t_1 = 8\ s$

$t_2 = 6\ s$

$l_1 = (v_1 + v_2) * t_1 = 45 * 8 = 340\ m$

$l_2 = (v_1 + v_2) * t_2 = 45 * 5 = 225\ m$

Javob: $l_1 = 340\ m,\quad l_2 = 225\ m$

2- Masala

Berilgan

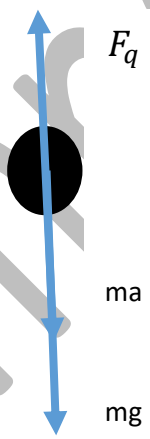
h=55 m

m=5 g

ϑ=10 m/s

g=10 $\frac{m}{s^2}$

F_q =?



Nyutonning 2- qonuniga ko'ra

$$ma = mg - F_q$$

$v = \sqrt{2ah}$ $\qquad a = \frac{v^2}{2h}$

$$m\frac{v^2}{2h} = mg - F_q$$

$$F_q = mg - m\frac{v^2}{2h}$$

$$F_q = 5 * 10^{-2} - 5 * 10^{-3} * \frac{100}{110} = 5 * 10^{-3} * \frac{100}{11} = 45{,}45 * 10^{-3}N = 45{,}45mN$$

Javob: $F_q = 45{,}45mN$

@olimpdep

3- Masala

Berilgan

$\Delta t = 0{,}1\ s$ $\quad t_1 = 0{,}7s$

$t_2 = 0{,}5s$

$$\Delta h = \frac{g{t_1}^2}{2} - \frac{g{t_2}^2}{2} = \frac{g}{2}({t_1}^2 - {t_2}^2)$$

$$\Delta h = 5(0{,}49 - 0{,}25) = 1{,}2\ m$$

Javob: $\Delta h = 1{,}2\ m$

4-**Masala**

Berilgan

$mg = 2{,}5\ N$ $\quad mg = \rho_j V g$

$mg - F_A = 2{,}25\ N$ $\quad mg - F_A = \rho_j V g - \rho_s V g = (\rho_j - \rho_s) V g$

$\rho =?$

$$\frac{\rho_j V g}{(\rho_j - \rho_s) V g} = \frac{2{,}5}{2{,}25} = \frac{10}{9}$$

$$9\rho_j = 10\ (\rho_j - \rho_s)$$

$$\rho_j = 10\rho_s = 10\ \frac{gr}{sm^3}$$

$$Javob: \rho_j = 10\ \frac{gr}{sm^3}$$

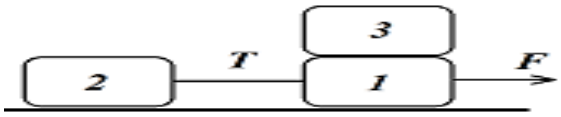
@olimpdep

5- Masala

Berilgan

$m_1 = m_2 = m$,

$F = const$, $\mu = 0$



Nyutonning ikkinchi qonuniga ko'ra

$3ma = F - 3\mu mg$

$3ma = F$

$2ma = F - T$

$T = F - 2ma = ma$

Uchinchi yuk ikkinchi yukni ustiga qo'yilsa

$$3ma_1 = F - 3\mu mg$$

$$3ma_1 = F$$

$$a_1 = a$$

$$T_1 = F - ma$$

$$T_1 = F - ma = 2ma$$

$$\frac{T_1}{T} = \frac{2ma}{ma} = 2$$

$$T_1 = 2T$$

Javob: 2 marta ortadi

@olimpdep